

5V

R2
33R

R7
33R

R9
33R

DLED1
TLCR5800

DLED2
TLCR5800

Q1
BC337-40

R1
220R

R4
47K

C2
100uF, 15V

GND

SENODIO

1 2 3

The RP2040 GPIO pins can support a max of 50mA TOTAL. RP2040 runs at 3.3v

Each pin should not draw more than 16mA
 $3.3/0.016 = 206$
 Let's go higher and say 220R

The diagram shows three photodiodes, LDPD1, RDPD1, and FPD1, all of type BPW96B. They are connected in a common-emitter configuration. The anodes (1) of all three photodiodes are connected to a common Vcc supply. The cathodes (2) are connected to a common node through resistors R11, R13, and R15, all with a value of 2k2. The nodes are labeled SEN0AIP, SEN2AIP, and SEN1AIP. The common node is connected to GND.

Diagram illustrating the Board Connector to Mezzanine connection for J1 (Conn_01x16).

Pin 1: PWR_FLAG

Pin 2: 5V

Pin 3: GND

Pin 4: Vcc

Pin 5: SEN0DIO

Pin 6: SEN1DIO

Pin 7: SEN2DIO

Pin 8: SEN3DIO

Pin 9: SEN4DIO

Pin 10: SEN0AIP

Pin 11: SEN1AIP

Pin 12: SEN2AIP

Pins 13, 14, 15, and 16 are marked with an 'X'.

The diagram illustrates the PCB layout for the power supply section, divided into two functional areas:

- Mounting Holes:** Two circular mounting holes are shown, each with a diameter of 3.2mm. Below each hole is a ground symbol (a triangle pointing down) labeled "GND".
- Power Smoothing:** This section is enclosed in a dashed blue border. It shows a vertical green line representing a trace. At the top, the trace is connected to a "VCC" pad. A capacitor, labeled "C3 330uF, 16v", is connected in parallel across the trace. The bottom of the trace is connected to a ground symbol labeled "GND".

Front Circuit

The RP2040 GPIO pins can support a max of 50mA TOTAL. RP2040 runs at 3.3v

Each pin should not draw more than 16mA
 $3.3/0.016 = 206$

Let's go higher and say 220R

5V

R5 33R

R8 33R

FLED1 TLCR5800

Q2 BC337-40

R6 47K

R3 220R

SEN1DIO

C1 100uF, 15V

GND

The diagram illustrates the connection of three indicator LEDs to a microcontroller. Each LED (D1, D2, D3) is connected to a specific microcontroller pin (SEN2DIO, SEN3DIO, SEN4DIO) through a 1KΩ resistor (R10, R12, R14). The other end of each LED is connected to ground (GND).

SEN0DIO	– “Diag LED” on PCB. Set High to turn on the diagonal sensor LED’s
SEN1DIO	– “Frnt LED” on PCB. Set High to turn on the front sensor LED
SEN2DIO	– “Left LED” on PCB. Set High to turn on the left Indicator LED. D1
SEN3DIO	– “Mid LED” on PCB. Set High to turn on the middle Indicator LED. D2
SEN4DIO	– “Right LED” on PCB. Set High to turn on the right Indicator LED. D3
SEN0AIP	– “Left ADC” on PCB. Read via ADC pin to get value from the left photodiode. LDPD1
SEN1AIP	– “Front ADC” on PCB. Read via ADC pin to get value from the front photodiode. FPD1
SEN2AIP	– “Right ADC” on PCB. Read via ADC pin to get value from the left photodiode. RDPD1

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